XUCHENG YU

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Trustworthy AI • LLM Safety and Reliability • Agentic and Multi-Agent Systems

RESEARCH INTERESTS

Trustworthy foundation models, with a focus on adversarial robustness, auditable reasoning and evaluation, and reliable agentic and multi-agent decision-making under uncertainty.

EDUCATION

University of Illinois Urbana-Champaign (UIUC) Urbana-Champaign, IL | Aug. 2024 - Dec. 2025
Master of Engineering in Electrical and Computer Engineering (ECE) | GPA: 4.0/4.0

Guangdong Technion - Israel Institute of Technology Shantou, China | Sep. 2020 - Jul. 2024
Dual degrees awarded by GTIIT and the Technion - Israel Institute of Technology
B.Sc. in Mathematics with Computer Science | GPA: 90.8/100, 3.7/4.0 | Dean's List, 2022-2023, 2023-2024

PUBLICATIONS AND MANUSCRIPTS

- **Xucheng Yu**, Haibo Jin, Huimin Zeng, and Haohan Wang. *SCI-Defense: Detecting Persuasion-Structured Manipulation of LLM-Based Ranking in Generative Engines.* arXiv:2605.21948. Submitted to The Web Conference 2027 (WWW 2027).
- Han Wang, Yifan Sun, Brian Ko, Mann Talati, Jiawen Gong, Zimeng Li, Naicheng Yu, **Xucheng Yu**, Wei Shen, Vedant Jolly, and Huan Zhang. *MonitorBench: A Comprehensive Benchmark for Chain-of-Thought Monitorability in Large Language Models.* arXiv:2603.28590. Accepted at COLM 2026.
- **Xucheng Yu**, Emily Knox, and Haohan Wang. *Understanding Content Moderation in Large Language Models through Restricted Books: From Refusal to Warning.* arXiv:2608.11806. Accepted at AIES 2026.
- Haibo Jin, Suijin Wang, **Xucheng Yu**, Haojing Luo, and Haohan Wang. *HEART: Harness Engineering in LLM Tool Use via Agent-Native Reusable Tool Primitives.* arXiv:2609.01736. Submitted to ICLR 2027.
- Ke Chen, **Xucheng Yu**, Yufei Zhou, and Haohan Wang. *Prompt Stability Matters: Evaluating and Optimizing Auto-Generated Prompt in General-Purpose Systems.* arXiv:2505.13546. Accepted at CPAL 2026.
- **Xucheng Yu**, Ruojia Tao, and Haohan Wang. *Closed-Loop Self-Improving Scientific Paper Generation: An Iterative Framework Combining AI Generation, Automated Review, and Targeted Optimization.* Manuscript.
- **Xucheng Yu**, Haibo Jin, Xiaoqi Wang, Runze Kong, and Haohan Wang. *Gap Detection: Mitigating Homogenization through Demand and Defensibility.* arXiv:2610.20162. Submitted to ICLR 2027.
- **Xucheng Yu**, Jiangping Chen, and Haohan Wang. *Where Does Rhetorical Distortion Live? A Multi-Platform, Multi-Language Measurement of Influencer Content.* arXiv:2610.26412. Submitted to The Web Conference 2027 (WWW 2027).
- Xiaoqi Wang, **Xucheng Yu**, Runze Kong, Haibo Jin, Huimin Zeng, Yifan Xu, and Haohan Wang. *GEO Survey and Unified Evaluation Benchmark.* Submitted to IEEE Transactions on Knowledge and Data Engineering (TKDE).

Submitted manuscripts are currently under peer review. Preprints are available on arXiv where applicable.

RESEARCH EXPERIENCE

Gap Detection: Mitigating Homogenization through Demand and Defensibility Urbana-Champaign, IL
Research Assistant | Advisor: Dr. Haohan Wang, Assistant Professor of ECE at UIUC Jul. 2026 - Oct. 2026

- Formulated a unified Gap Detection framework connecting three forms of AI content homogenization: LLM output diversity, GEO market positioning, and synthetic training-data collapse, using Demand x Defensibility optimization in embedding space.

- Developed theoretical and empirical analyses including the Unmet-Demand Theorem and Nash product objective; reduced LLM response collision by 12.8% on curated topics and 10.8% across hundreds of INFINITY-CHAT model responses.

Rhetorical Distortion across Platforms and Languages Urbana-Champaign, IL
Research Assistant | Advisor: Dr. Haohan Wang Jul. 2026 - Oct. 2026

- Measured rhetorical distortion across 13,060 English and Chinese posts from 330 accounts on six platforms; found platform more strongly associated with distortion than content category (partial η^2 0.49 vs. 0.09), with X/Twitter replies carrying 3.5× the distortion of the tweets they answer.
- Built a three-tier pipeline combining bilingual lexical rules, negative-pattern filtering, and GPT-4o-mini verification for five distortion dimensions, calling the LLM on 14% of posts; validated on 198 posts labeled by two human annotators (inter-annotator Cohen's kappa = 0.76).

GEO Survey and Unified Evaluation Benchmark Urbana-Champaign, IL
Research Assistant | Advisor: Dr. Haohan Wang May 2026 - Oct. 2026

- Systematically characterized adversarial manipulation of LLM ranking systems across 20 generative engine optimization (GEO) pipelines, establishing a benchmark foundation for evaluating trustworthiness under deployment-time attacks.
- Compared AutoGEO, MAGEO, AgenticGEO, and E-GEO to identify algorithmic assumptions and new optimization and defense directions.

SCI-Defense: Persuasion-Structured Ranking Manipulation Urbana-Champaign, IL
Research Assistant | Advisor: Dr. Haohan Wang Jan. 2026 - May 2026

- Designed a defense in which an LLM judge scores five persuasion devices, complemented by perplexity and relevance gates; detected 89.6% of reasoning attacks and 96.2% of review attacks at a 1.2% false-positive rate across 1,152 CORE attacks on Amazon and MS MARCO.
- Blocked 96.1% of attacks that pushed their target into the top three while displacing 0.6% of honest items, versus 4.0-4.5% for paraphrasing and truncation; mapped failure modes with six black-box attacks, including relevance inflation.

HEART: Reliable Multi-Agent Tool Use Urbana-Champaign, IL
Research Assistant | Advisor: Dr. Haohan Wang Jan. 2026 - May 2026

- Designed Tool Primitives and HEART's Planner-Router-Verifier workflow; built ToolFace, a repository of 25,519 functions supporting dynamic retrieval, nested calls, and feedback-driven recovery.
- Evaluated on five diverse benchmarks, demonstrating a 10% improvement over SFT baselines and 6% over frontier commercial models, while reducing token consumption by up to 85%, highlighting sample efficiency in multi-agent tool use.

Content Moderation through Restricted Books Urbana-Champaign, IL
Research Assistant | Advisor: Dr. Haohan Wang Jan. 2026 - May 2026

- Executed a 40,800-pair empirical study across 400 books and 17 prompt designs, evaluating six frontier models (Claude Sonnet 4.5, GPT-4o, Gemini 2.5 Flash, DeepSeek V3, Qwen-Plus, Grok-4.1-Fast); uncovered a near-zero refusal rate (0.07%), revealing systematic vulnerabilities in cross-model safety alignment.
- Characterized the shift from refusal to warning-based moderation, including warning gaps of 8-15 percentage points and prompt-dependent gaps reaching 19 points.

MonitorBench: A Comprehensive Benchmark for Chain-of-Thought Monitorability in Large Language Models Urbana-Champaign, IL
Research Assistant | Advisor: Dr. Huan Zhang, Assistant Professor of ECE at UIUC Jan. 2026 - Apr. 2026

- Contributed to MonitorBench, a comprehensive benchmark evaluating chain-of-thought monitorability across 1,514 instances, 19 tasks, and seven categories, with standard, direct-concealment, and monitor-aware-evasion settings.
- Implemented the impossible-coding-task evaluation adapted from ImpossibleBench and engineered a parallel multi-model inference and verification pipeline for large-scale monitorability experiments.

Prompt Stability and Multi-Agent Systems for Data Science Urbana-Champaign, IL
Research Assistant | Advisor: Dr. Haohan Wang Feb. 2025 - Sep. 2025

- Established semantic stability as a key reliability criterion for auto-generated prompts, devising an optimization framework linking prompt consistency to system-level task success.

- Architected a full-stack multi-agent platform with editable and regenerable agent dialogues, reaching 95% task completion and reducing user intervention by 40%.

RESEARCH SYSTEMS & PLATFORMS

Generative Engine Optimization Platform Urbana-Champaign, IL
Research Engineer | TopCited.ai May 2026 - Present

- Developing GEO SaaS features for AI visibility auditing, content optimization, and SCI-Defense-based ranking-manipulation detection across ChatGPT, Gemini, and Perplexity.

Natural-Language Tool Orchestration Platform Urbana-Champaign, IL
Research Engineer | Heelo May 2026 - Present

- Engineering natural-language tool orchestration with dynamic routing, multi-turn clarification, browser-based integrations, and credential-isolated execution.

PROFESSIONAL EXPERIENCE

Donovan's Piano Room Urbana-Champaign, IL
Full-Stack Engineer Feb. 2026 - Jul. 2026

- Resolved authentication and session-management defects across the Next.js frontend, Node.js/TypeScript backend, and Redis infrastructure by redesigning token-validation logic and decoupling protected routes from short-lived access tokens.

Bosch Automotive Products (Suzhou) Co., Ltd. Suzhou, China
Software Engineering Intern Sep. 2023 - Oct. 2023

- Developed and deployed an IoT data-processing and machine-learning pipeline using Airflow, Pandas, TensorFlow, FastAPI, and C/Cython, reducing preprocessing time by 40%, improving model accuracy by 12%, lowering query latency by 30%, and accelerating inference by 10×.

TECHNICAL PROJECTS

Augmented Reality Interactive Modeling for Kohler Urbana-Champaign, IL | May 2025 - Aug. 2025

- Developed a real-time AR inspection pipeline using Python, Open3D, and Unity, achieving 0.1–0.2 mm precision and supporting ICP-based pose tracking for meshes with over 10 million vertices.

RLHF for Qwen2.5-7B-Instruct Urbana-Champaign, IL | Jan. 2025 - May 2025

- Implemented an end-to-end PPO/GAE/KL-penalty pipeline with LoRA; achieved 70.1% pairwise preference accuracy and a 64.7% win rate versus 33.0% for the baseline on Stack Exchange data.

Distributed Consensus and Fault Detection Urbana-Champaign, IL | Jan. 2025 - Apr. 2025

- Implemented Paxos and Raft over gRPC/Protobuf with 99.9% leader-election success and sub-200 ms election latency under network partitions; halved fault-detection time in simulated failures.

TEACHING EXPERIENCE

Peer Tutor & Grader Guangdong Technion - Israel Institute of Technology

- Tutored lower-year students in mathematics and computer science coursework, supporting concept clarification, problem-solving strategies, and adaptation to a rigorous Technion-aligned curriculum.
- Graded assignments for Calculus, Topology, and Probability, providing feedback on mathematical reasoning, proof structure, and solution clarity.

TECHNICAL SKILLS

Research Methods: LLM evaluation, benchmark design, adversarial testing, large-scale empirical analysis, RLHF, multi-agent system evaluation
Programming Languages: Python, C/C++, Go, Java, TypeScript/JavaScript
Frameworks & Technologies: PyTorch, vLLM, SGLang, LoRA, FastAPI, Docker, React, Node.js, gRPC/Protobuf